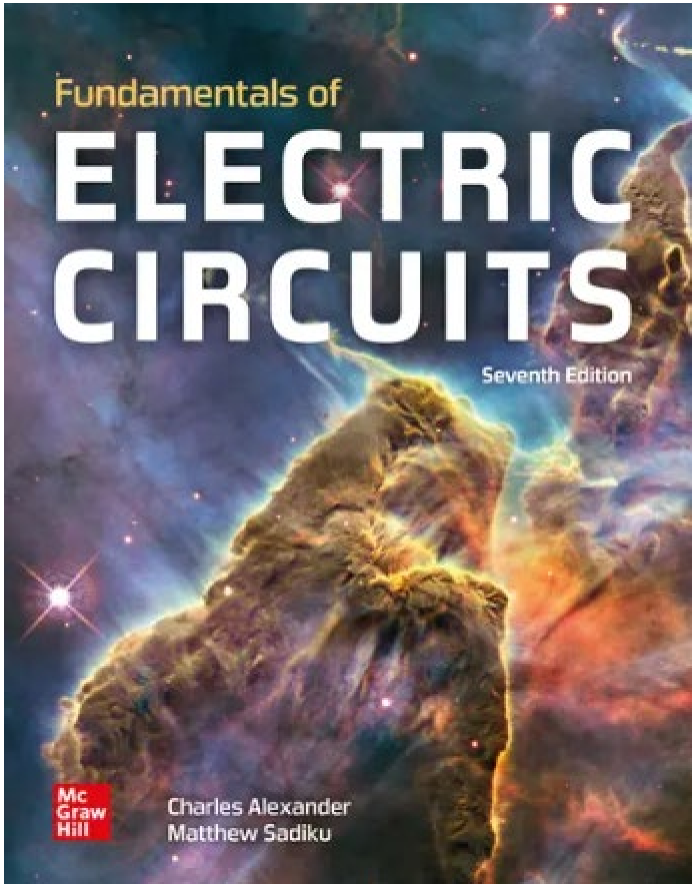
Electrical Science
ENGR 2303

Dr. Mohamed Bingabr
School of Engineering
University of Central Oklahoma

Course Materials

Textbook: "Fundamentals of Electric Circuits", by Alexander and Sadiku.



Course Website:

http://www.engineering.uco.edu/~mbingabr

Course Information

Instructor: Mohamed Bingabr, Prof. of Engineering

Office Location: HOH 221C

Phone: (405) 974 5718

Email: mbingabr@uco.edu

Course Meeting Time: MTWR 9:30 – 10:45 am

Course Meeting Location: HOH 205

Office Hours: MW 12:30 – 1:30pm & by appointment

Grading Policy and Distribution

Attendance	10 %
Homework	10 %
Quizzes	30 %
2 Tests	30 %
Final Exam	20 %

A: 90-100
B: 80-89
C: 70-79
D: 60-69
F: 0-59

Don't wait till the final day to ask me what you can do to pass the course.

Name	Q1	Q2	Q3	Q4	Q5	Q6	Q7	TQ	T1	T2	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	THW	Attd	Final	Total	Letter
								30%	15%	15%												10%	10%	20%	100%	Grade
								0.0														0			0.0	
								0.0														0			0.0	
								0.0														0			0.0	
								0.0														0			0.0	
								0.0														0			0.0	
								0.0														0			0.0	
								0.0														0			0.0	
								0.0														0			0.0	
								0.0														0			0.0	
								0.0														0			0.0	
								0.0														0			0.0	
Max Grade	20	20	20	20	20	20	20	30.0	100	100	20	20	20	20	20	20	20	20		20	20	10	10.0	100	100.0	
Average																										

Course Expectations & Conducts

- Spend a total of 8 hours a week.
- Ask for help for homework but don’t copy.
- You are allowed to have handwritten one page of formula sheet both sides for quizzes and tests.
- You can not use unallowed resources during quizzes and tests.
- Makeup test will be given for emergency.
- Please don’t use the phone or the internet during lectures.
- Be on time for the lecture and do your best to stay focus.

Course Topics

- Ohm's Law, Kirchhoff's Law Series and Parallel Resistor
- Nodal and Loop Analysis Techniques
- Operational Amplifiers, Op-Amp Models
  - Test 1

- Superposition, Thevenin's and Norton's Theorem
- Maximum Power Transfer
- Capacitor and Inductor RLC Circuit
- RC Operational Amplifier Circuits
  - Test 2

- First Order Circuit
- AC Steady-State Analysis
- Instantaneous and Average Power
- Variable-Frequency Network Analysis and Performance (Filters)
  - Final

To Do Well in This Course

- Focus on lectures & problems solved in the lecture.
- Do and understand the homework problems.
- Enjoy and reflect on the physical meaning of the equations and engineering concepts.
- Time management: study and do homework first then socialize and have fun.
- Use video resources in YouTube “Engineering with Bingabr”